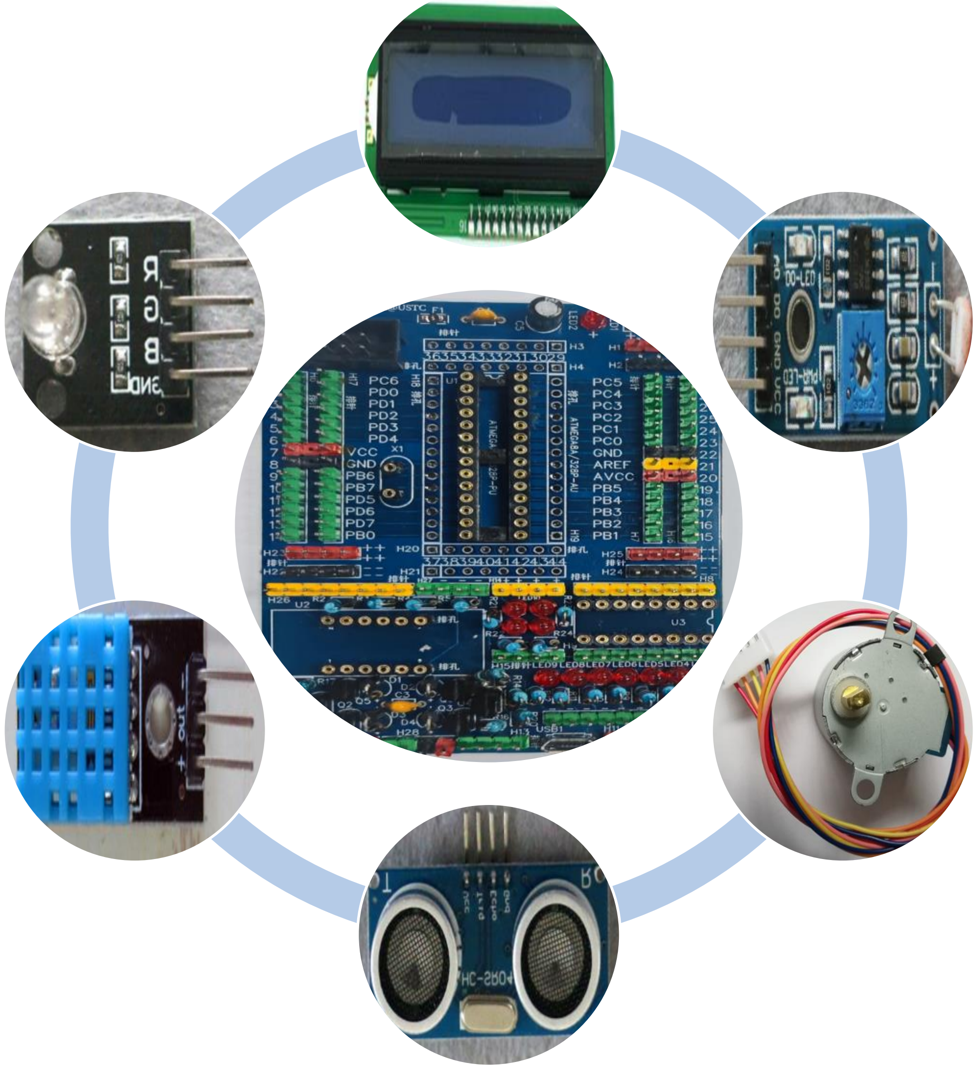
电子设计实践基础

电子系统与电子元器件



硬件

学习、办公、娱乐，
设计、制作、研究…

✚

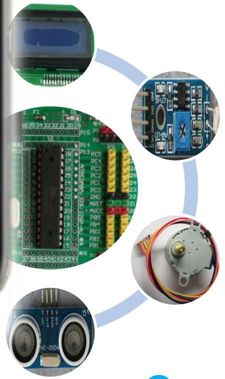
Linux、Windows、
macOS…；Apps

Android、windows、
iOS…；APPs

✚

通信、拍照、购物，
商务、生活、娱乐…

软件



软件（Software）

OS(Operating System) + Apps(Applications)

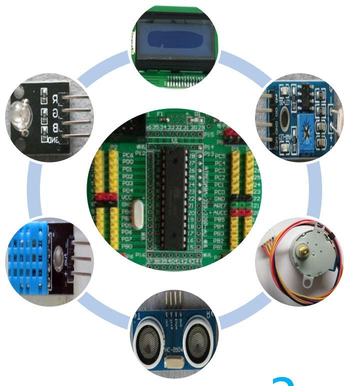
软件

一定方式+硬件指令+组织+特定的任务

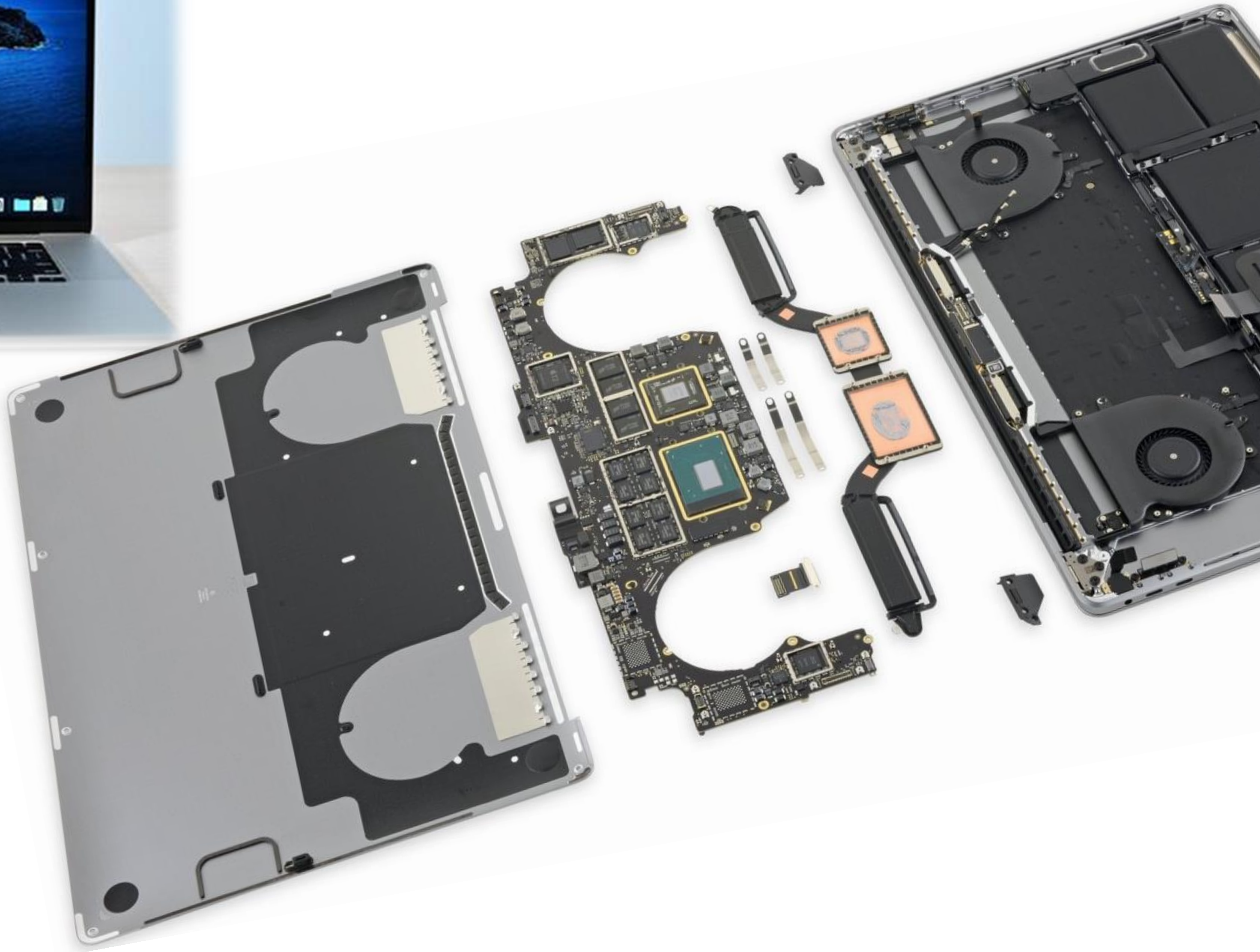
算法

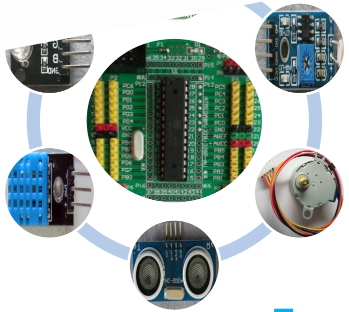
ICs

OSs/Apps




HUAWEI





硬件（Hardware）

Central Processing Unit

CPU/CU(南北桥)+内存+主板+功能模块+机械

SoC

Control Unit

计算/控制

指令/程序

硬件

互连

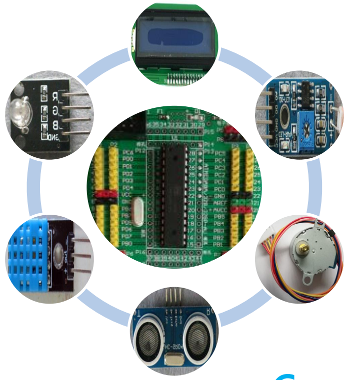
IO/外设

安装/组合/成型

物理设备/装置

电阻 (R)
电感 (L)
电容 (C)
晶体管 (T)
集成电路 (IC)

电子、机械、光电、电磁等



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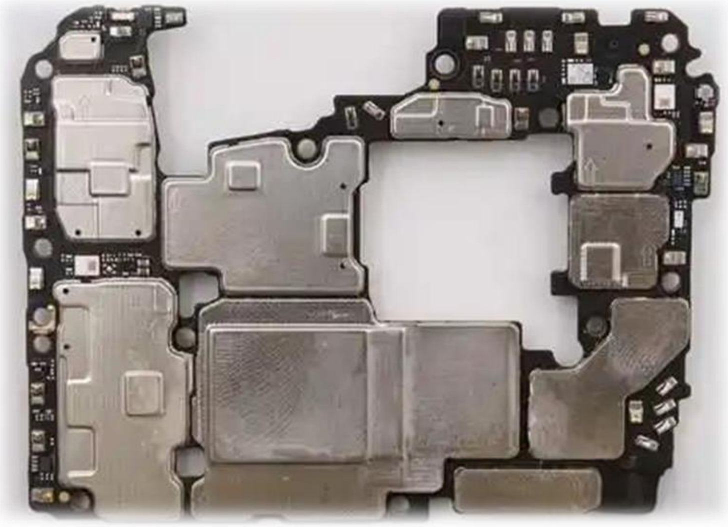
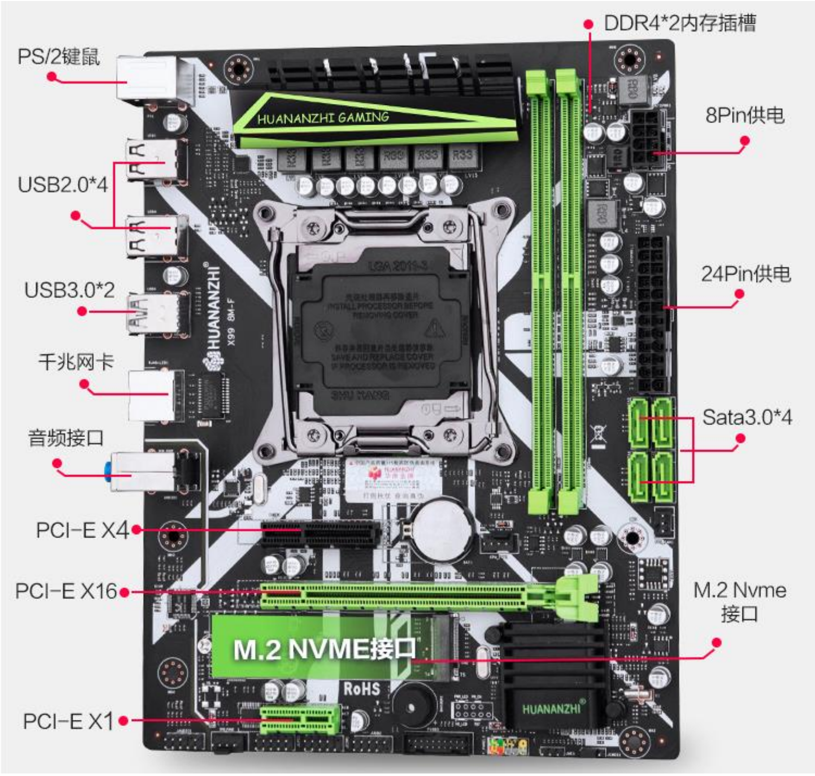
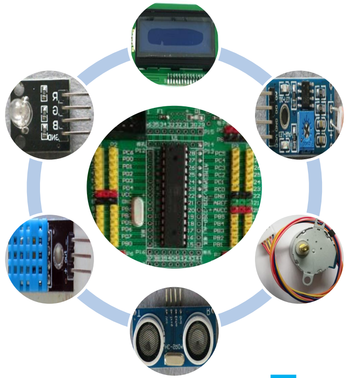
常用电子元器件的认识与使用



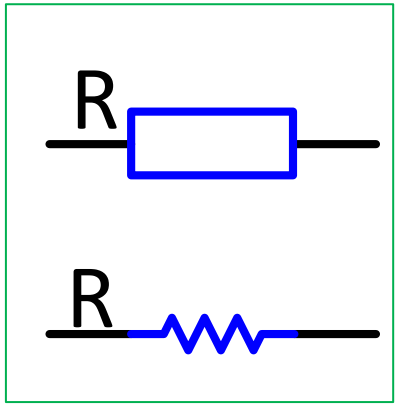
- 电阻
- 电容
- 电感

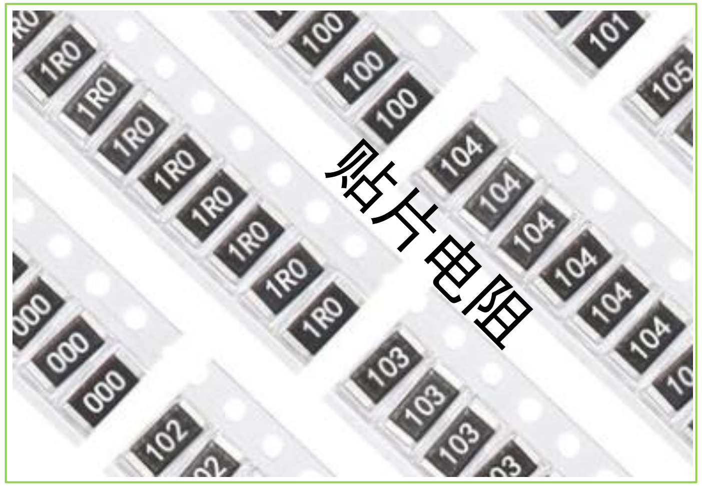


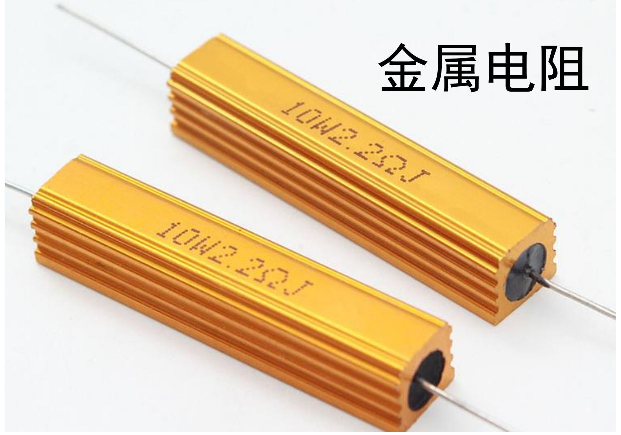
- 保险丝
- 二极管
- 三极管
- 集成电路芯片

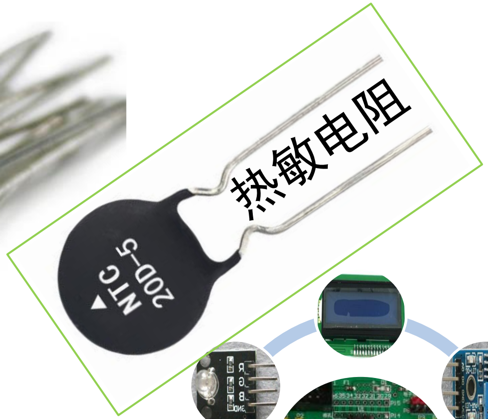


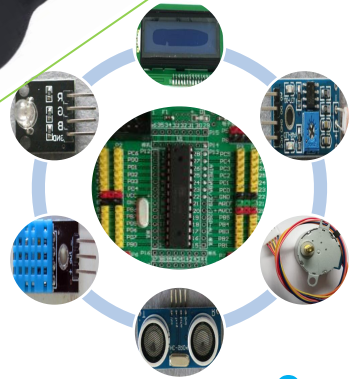
电阻1：认识电阻（R：Ω）











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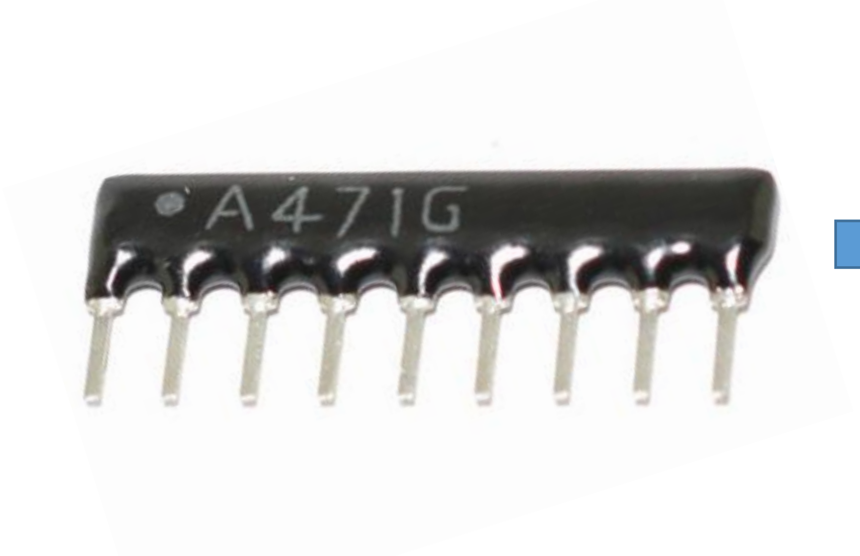
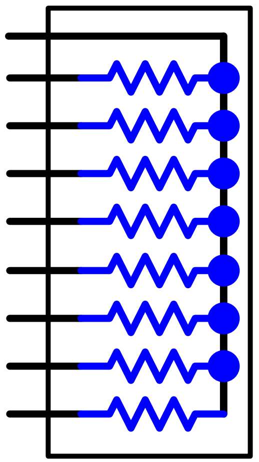
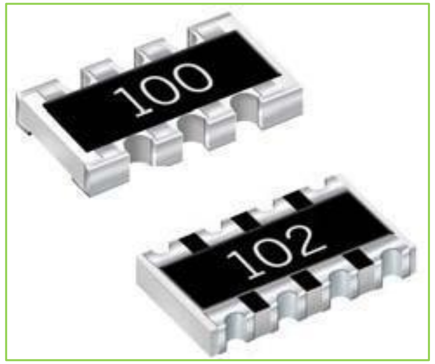
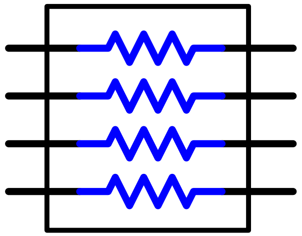
电阻2：认识排阻









- 标识与电阻值：

  $100=10*10^0=10\Omega$

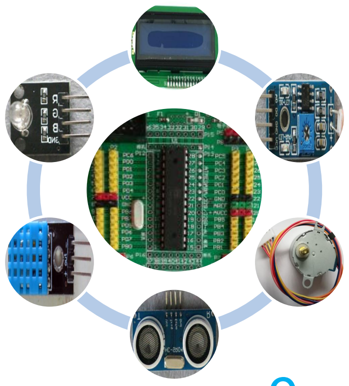
  $102=10*10^2=1000\ \Omega$

A471G：

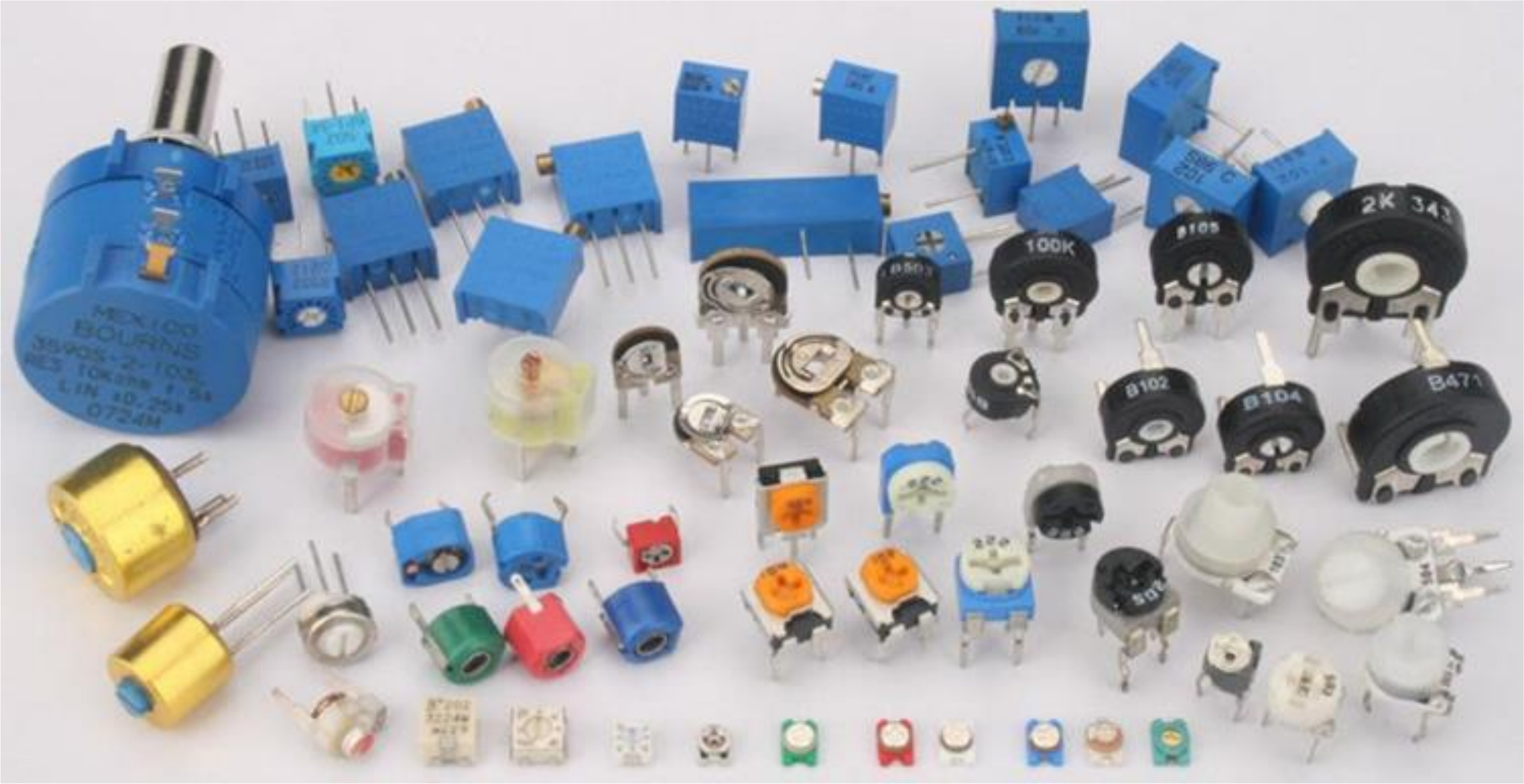
A公共端

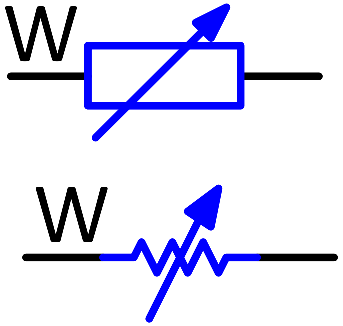
$471=47*10^1=470$

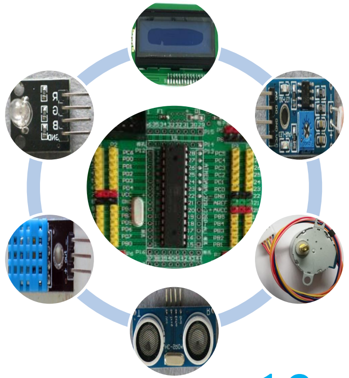
G=2%（误差）



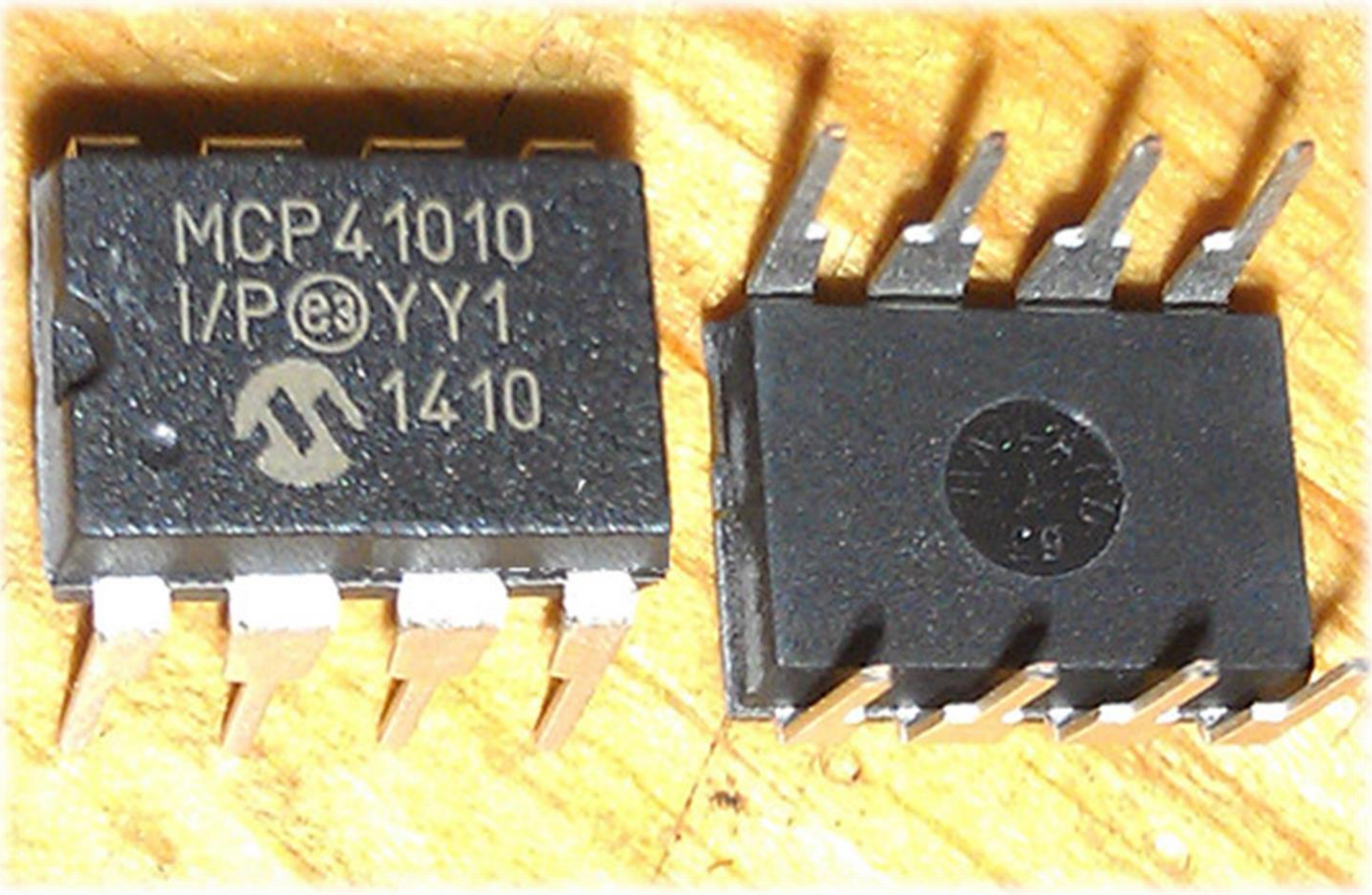
电阻3：可变电阻



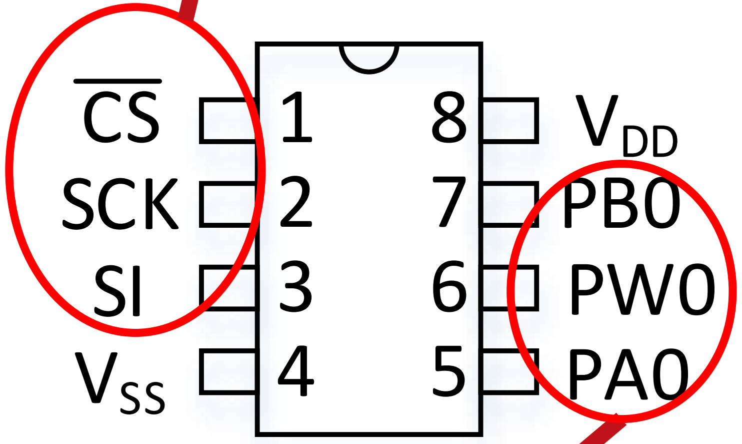




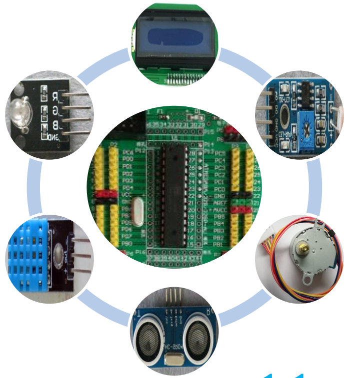
电阻4：数字电位器



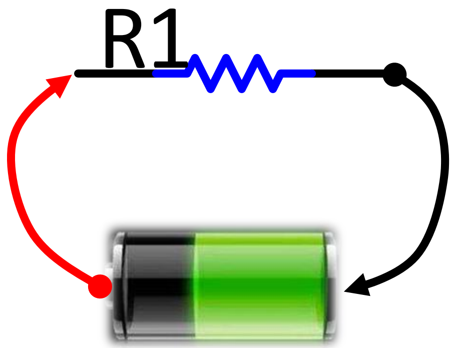
SPI数字控制接口



可变电阻管脚



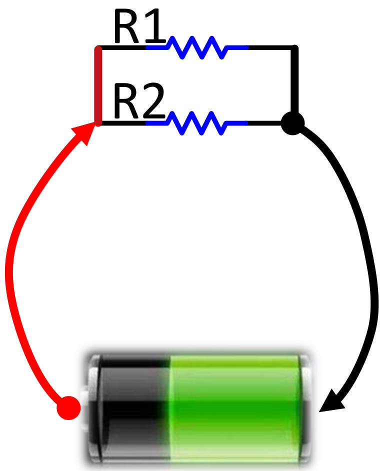
电阻的作用：阻碍电流流动

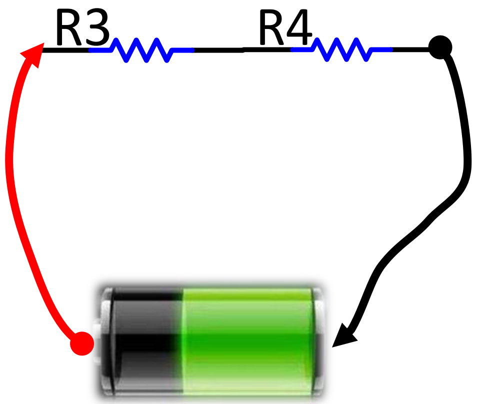


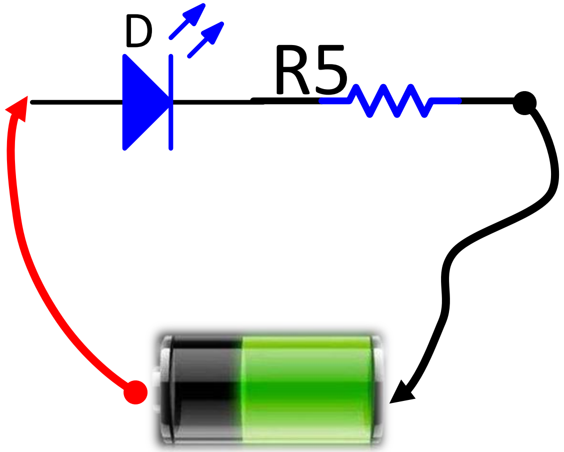
阻碍电流的流动

分流、分压、限流...

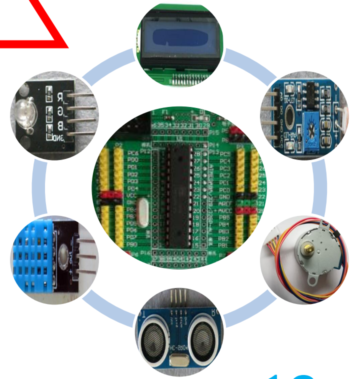
端接、阻抗匹配、...



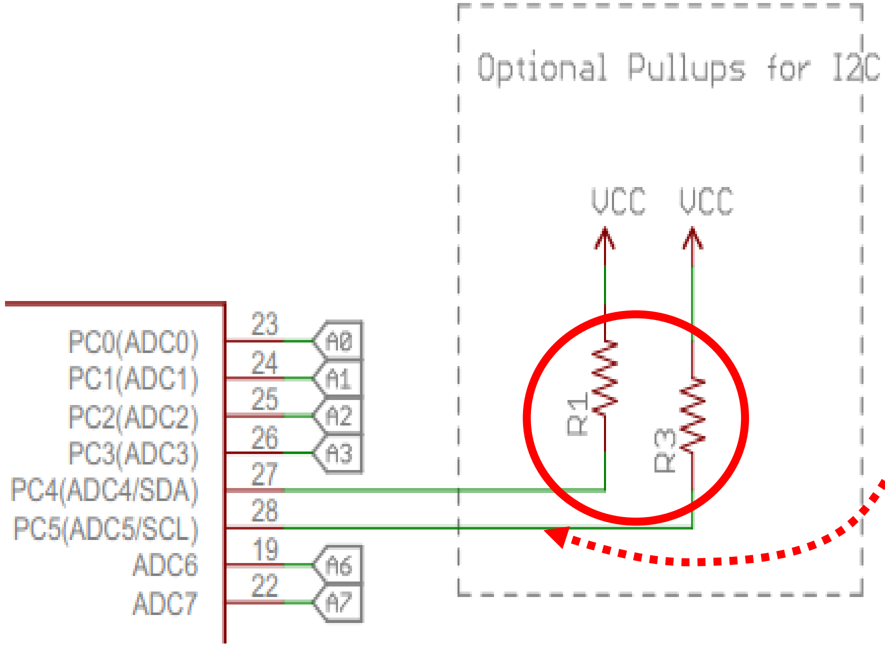




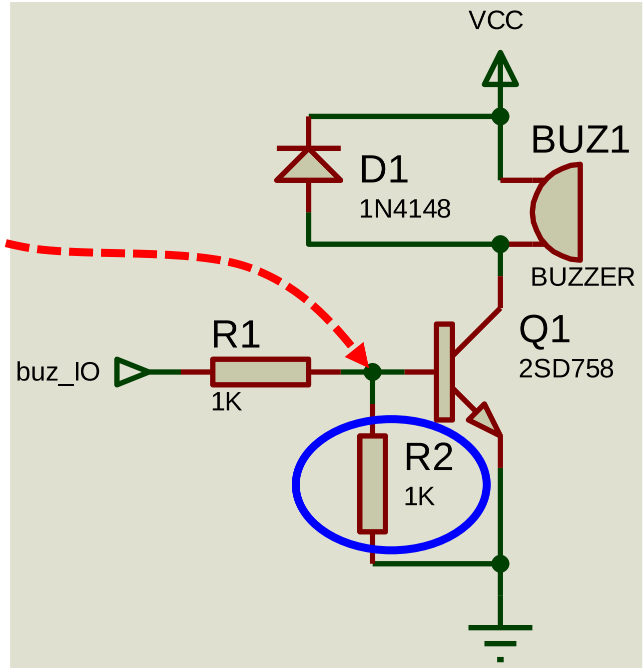
交流信号

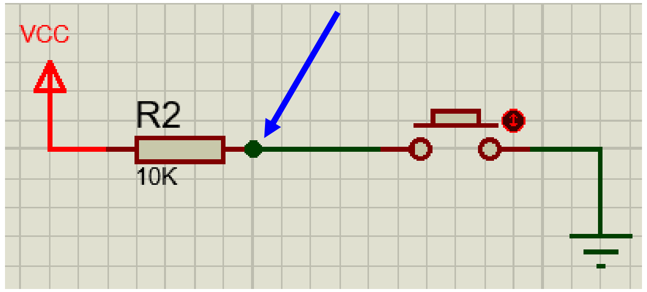


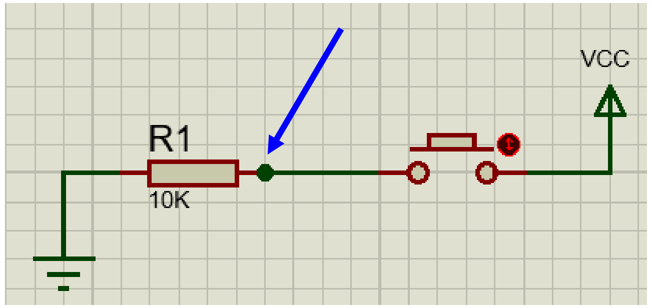
电阻的应用1：上拉、下拉



常用来提供确定的电平状态

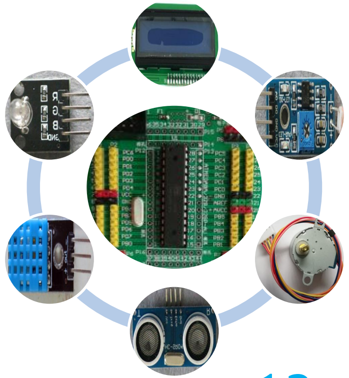






下拉电阻：电阻一端连到地(-)

上拉电阻：电阻一端连到电源(+)



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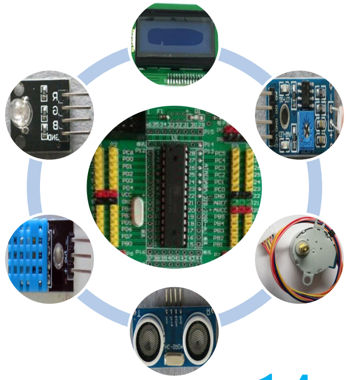
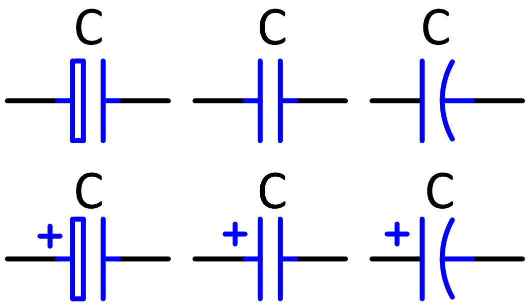
电阻的应用2：0欧姆电阻

0欧姆电阻

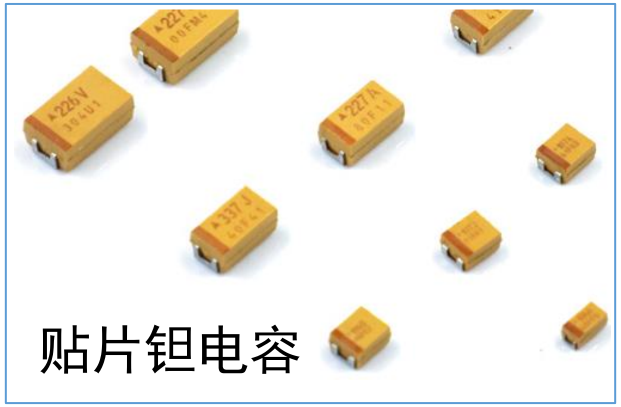
- 飞线
- 跳线
- 测试点
- 配置
- ‘熔丝’
- 替代



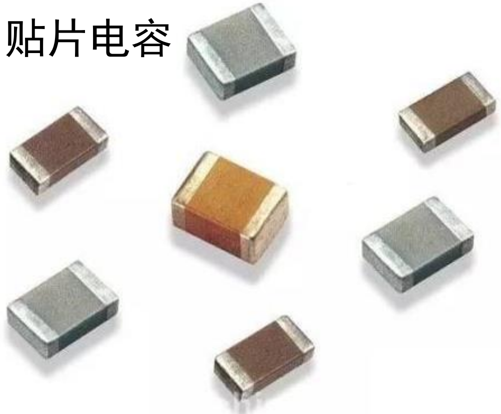
电容1：认识电容(C：F)



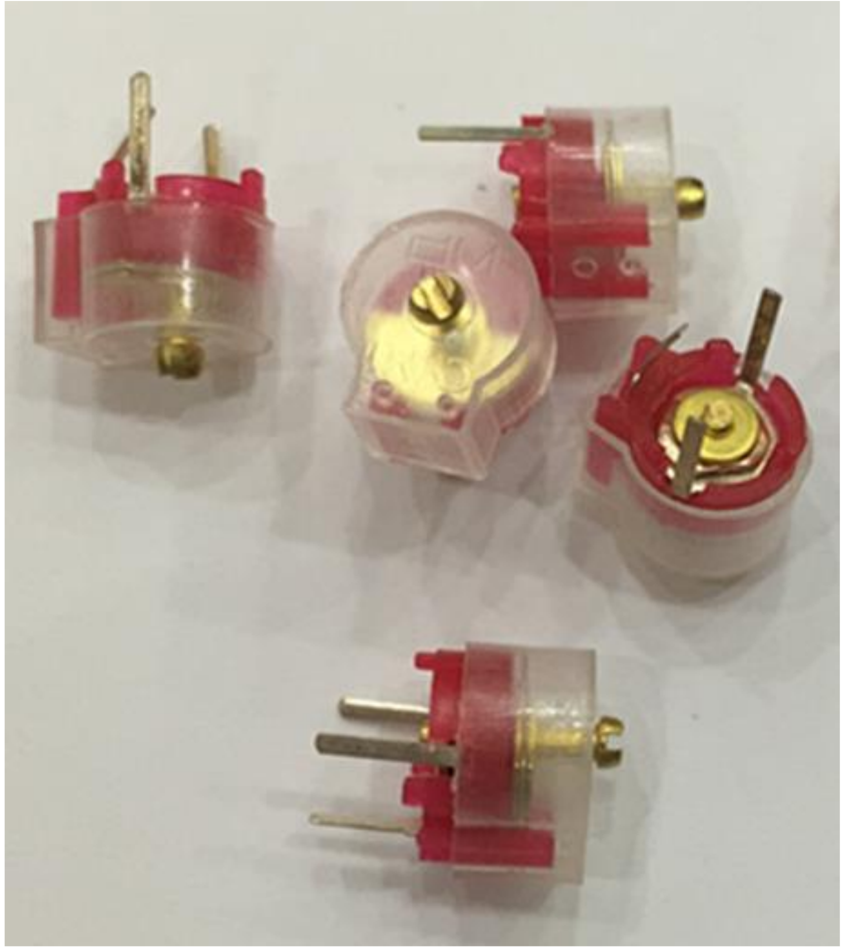
瓷片电容

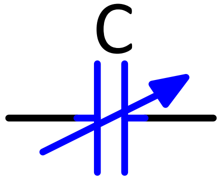


贴片电容

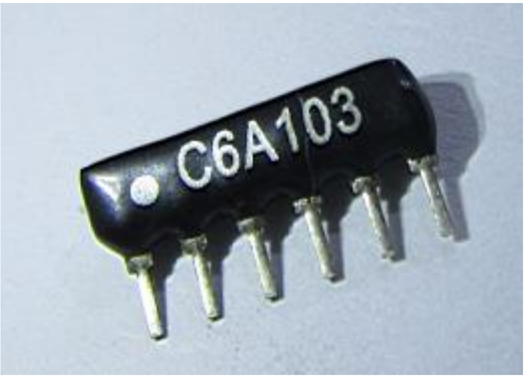


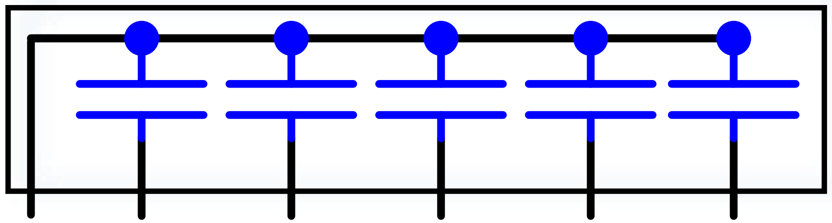
电容2：可变电容





电容3：排容



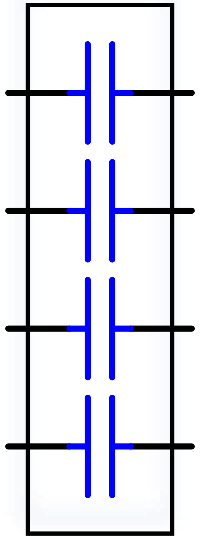


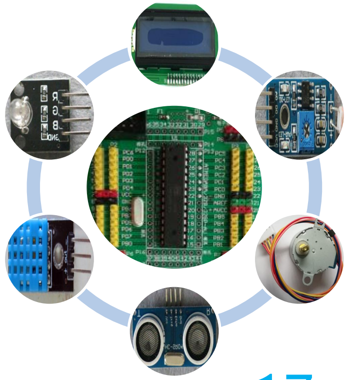
- 标识与电容值：

C6A103：C-电容，6管脚数

A公共端

$103=10*10^3=10^4$pF





电容的作用

存储电荷的器件

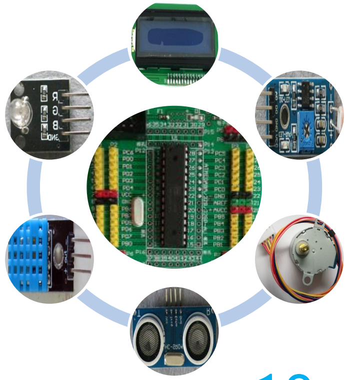
↓

隔直流、通交流

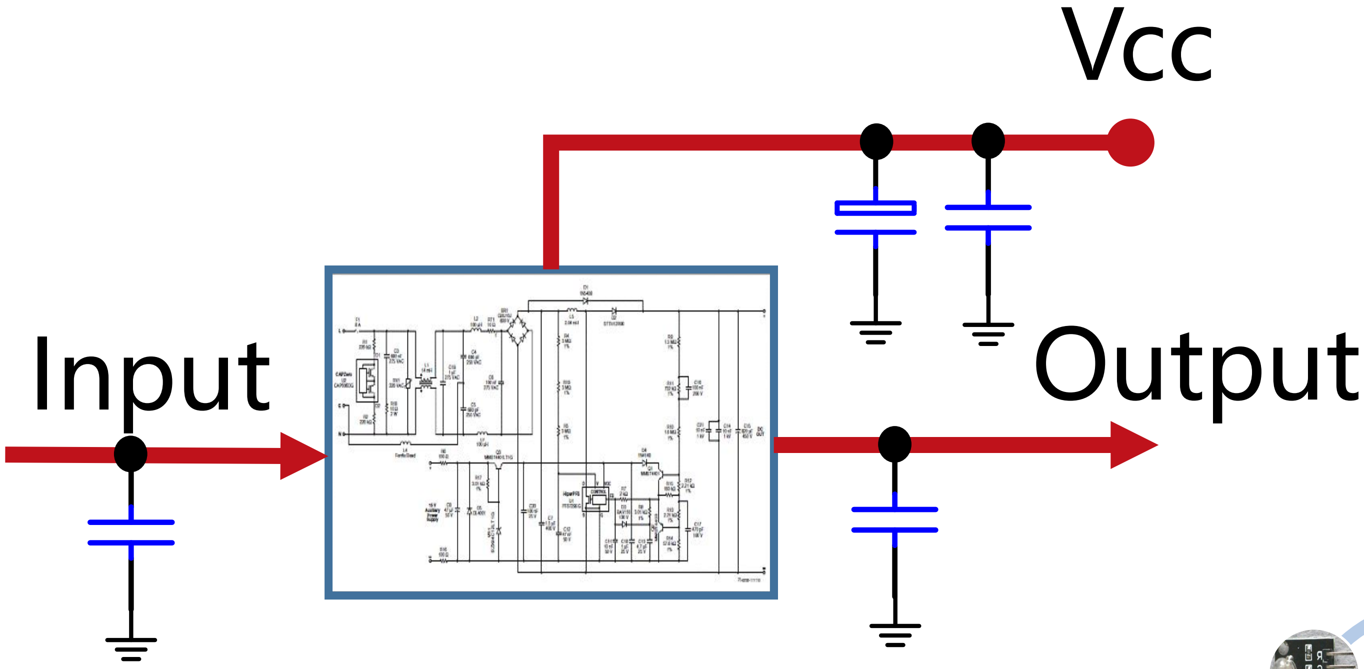
↙ ↘

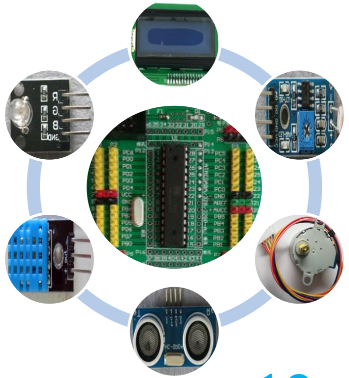
旁路、去耦、滤波

谐振、耦合、储能

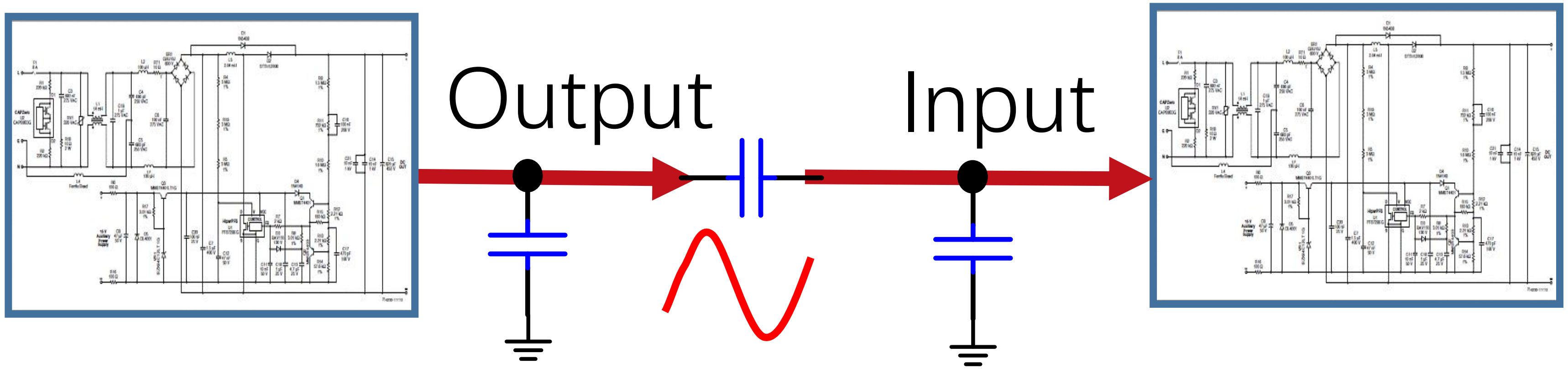


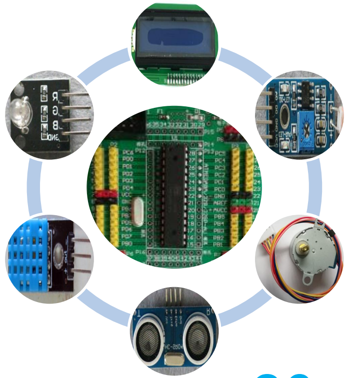
电容的应用1：旁路、去耦、滤波



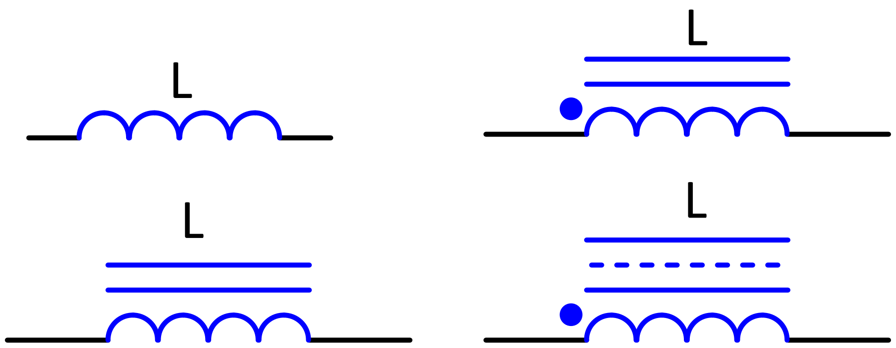


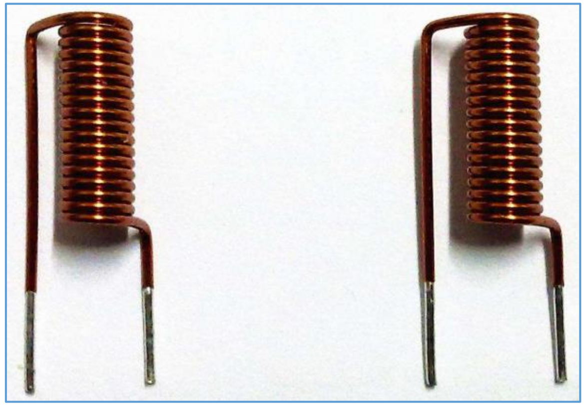
电容的应用2：耦合

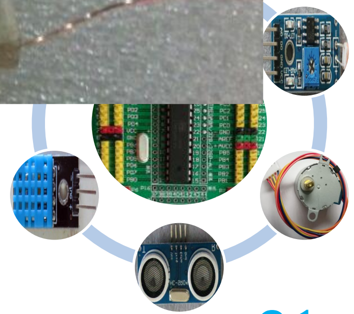




电感1：认识电感







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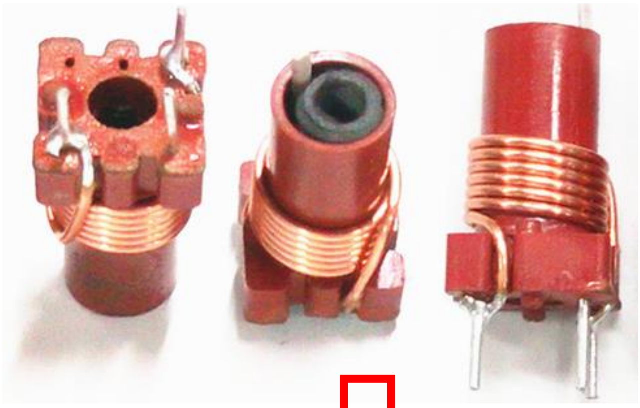
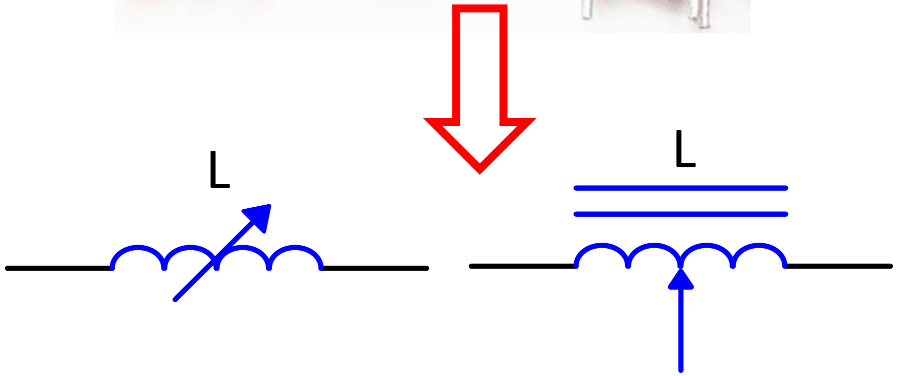
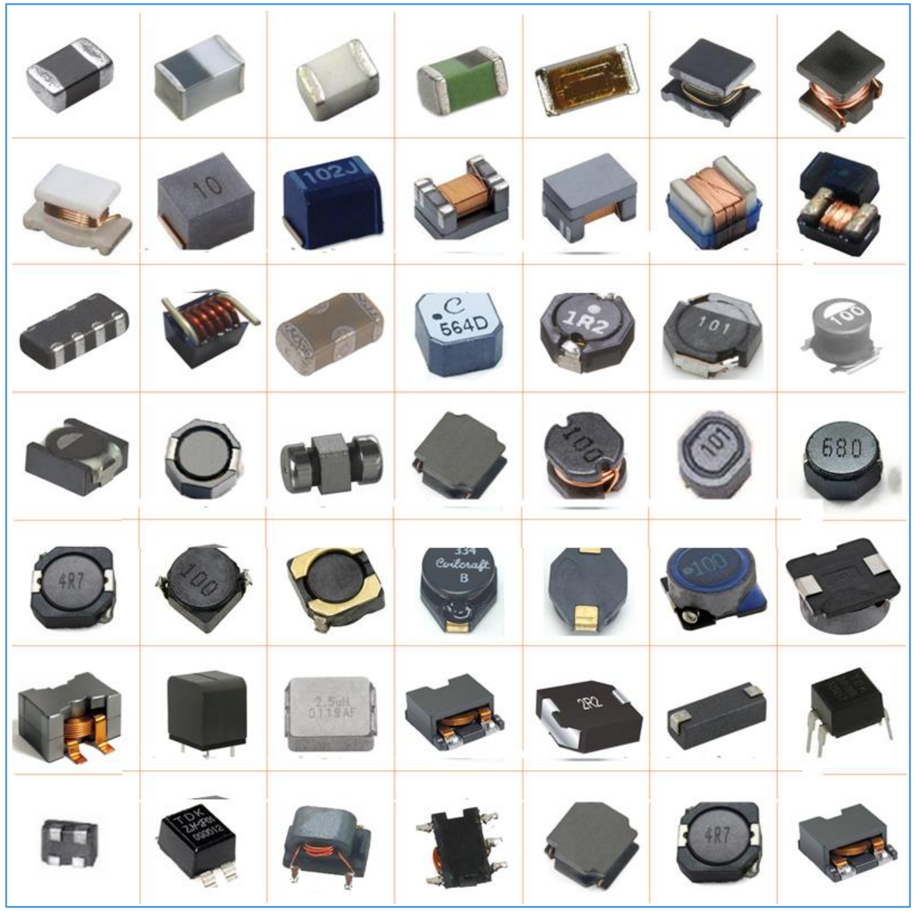
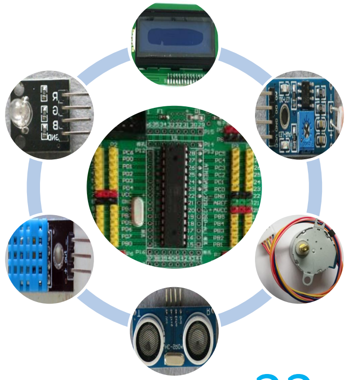
电感2：可调电感、贴片电感

可变电感器









电感的作用

把电能转化为磁能并储存

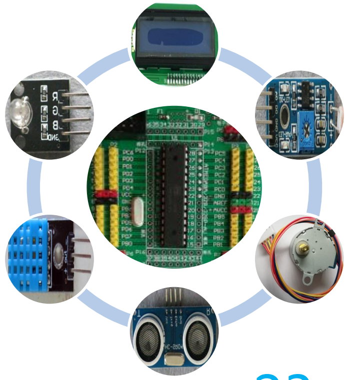
↓

通直流、阻交流

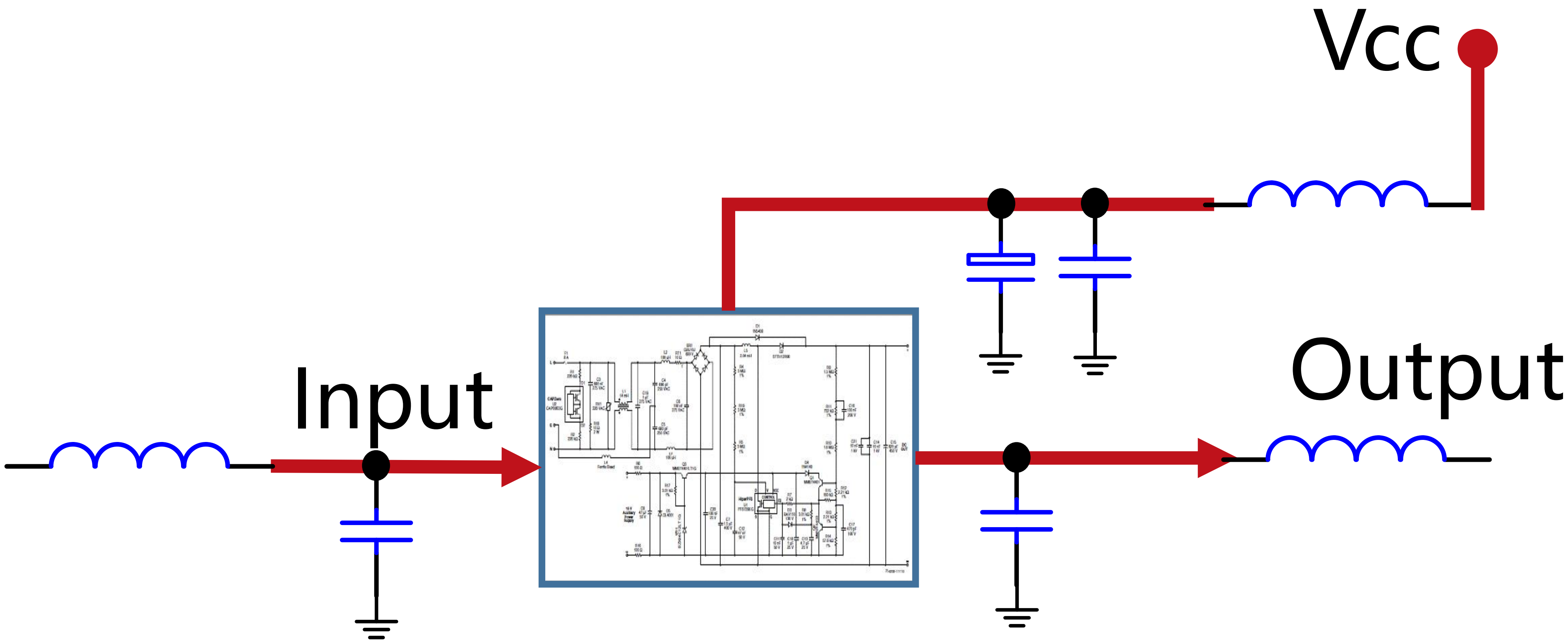
↙ ↘

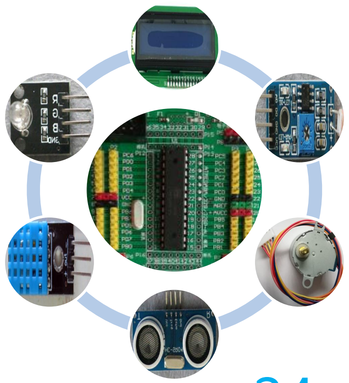
滤波、陷波、延迟

振荡、稳流、升压

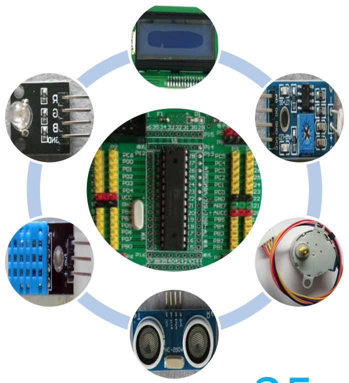


电感的应用1：LC滤波、电源滤波等





电感的应用2



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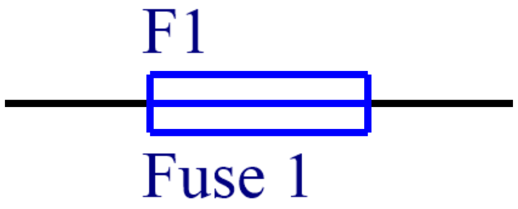
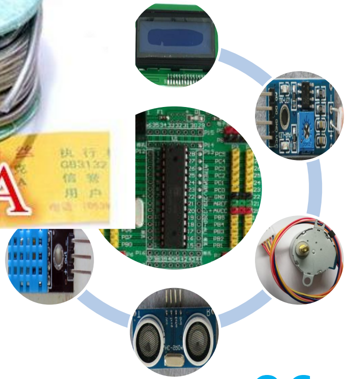
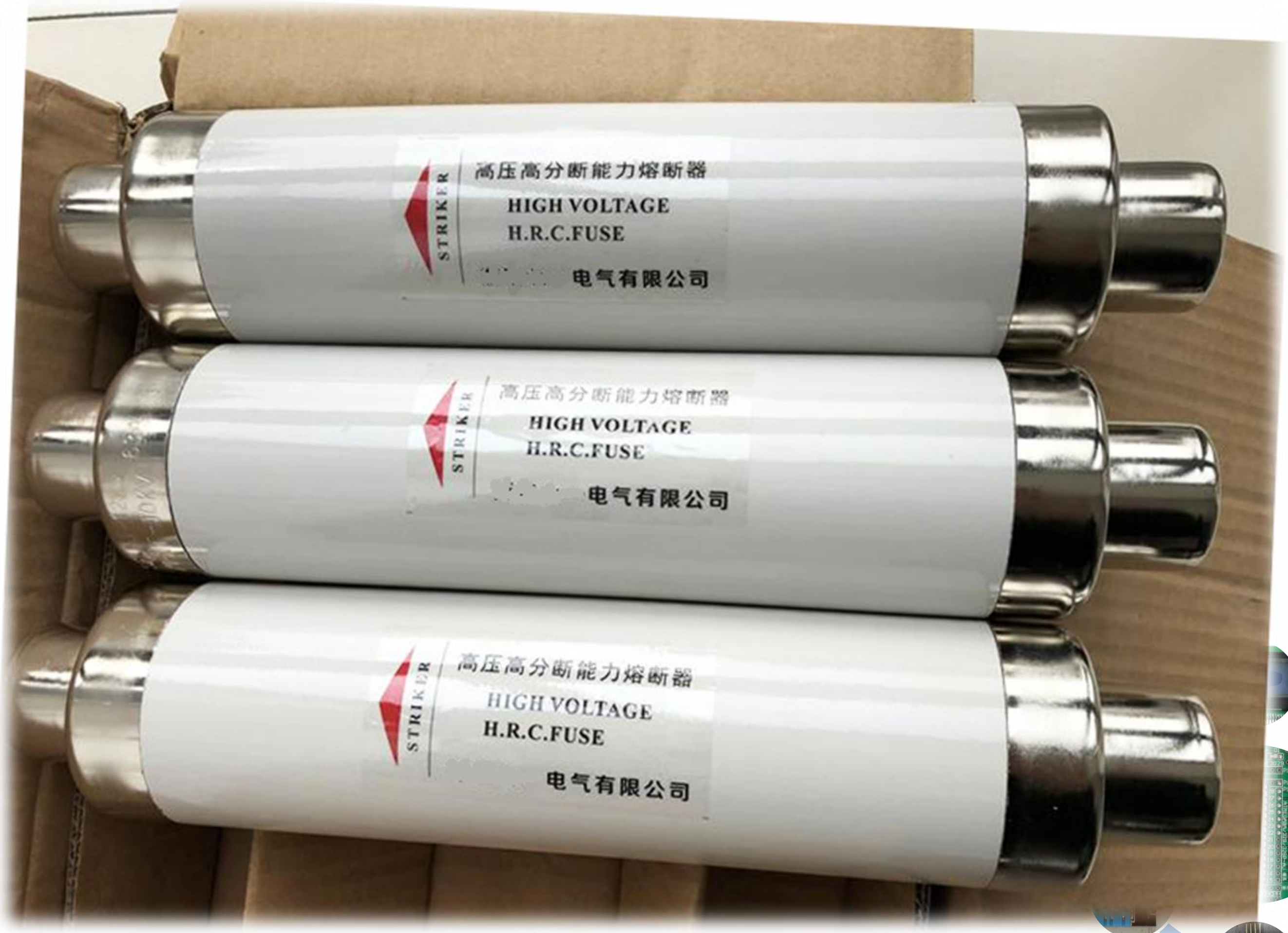
保险丝1：认识保险丝1

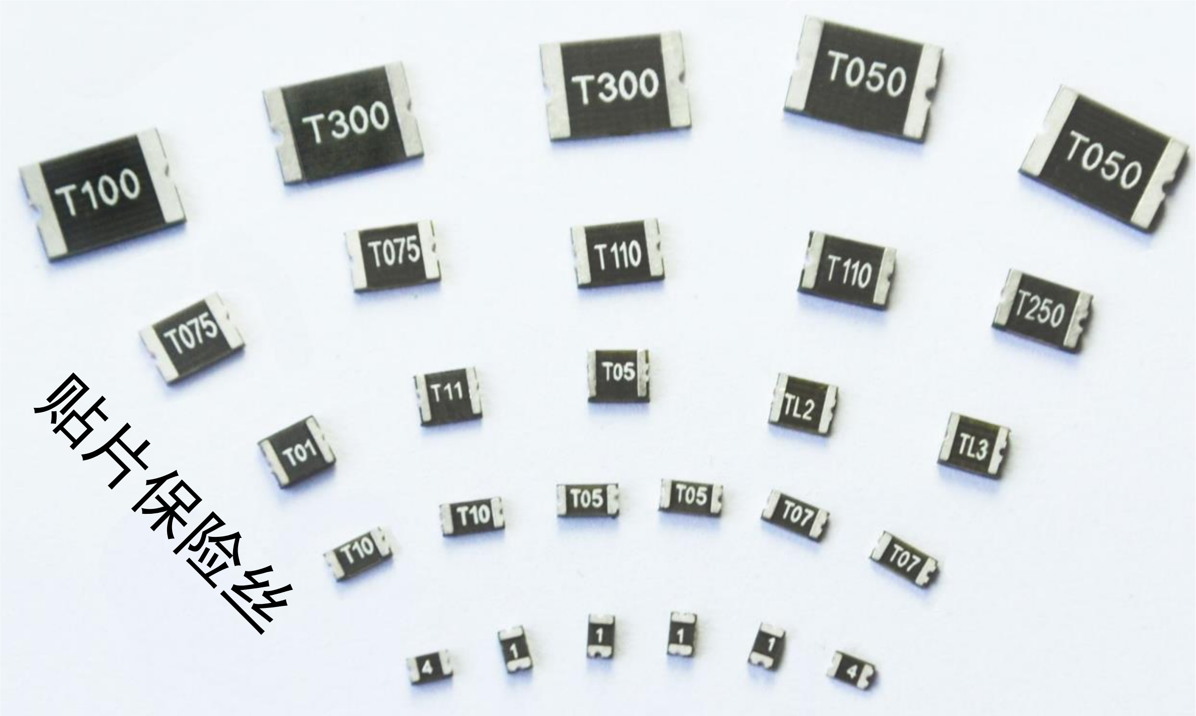




保险丝1：认识保险丝2



保险丝1：认识保险丝3


T100
T300
T300
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T050
T075
T110
T110
T075
T250
T11
T05
TL2
T01
TL3
T10
T05
T05
T07
T10
T07
贴片保险丝


2A 3A 5A 7.5A 10A
15A 20A 25A 30A 35
5A 7.5A 10A 15A 20A 25A
5A 7.5A 25A
汽车保险丝

保险丝的作用

过载保护

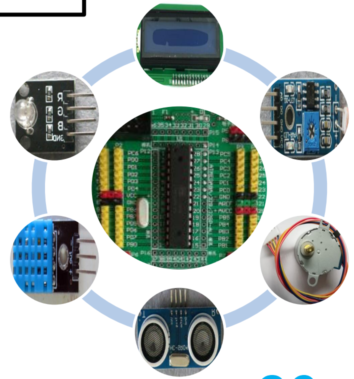
↓

自身熔断/切断以断开电路电源

↙ ↘

保护电路安全运行

保护设备、降低损失



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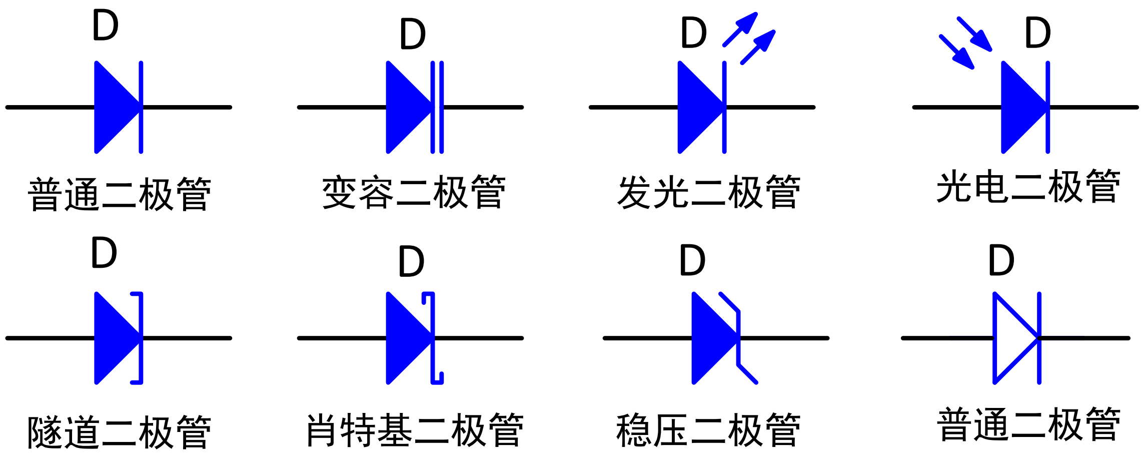
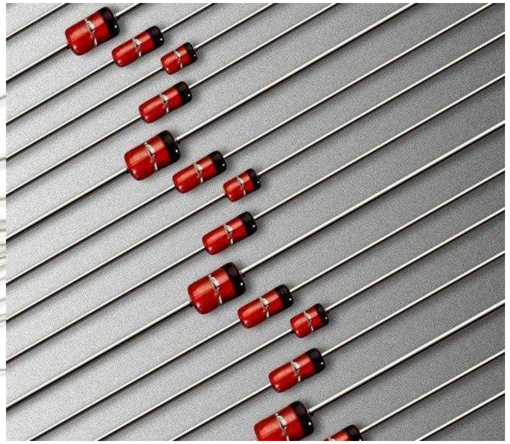
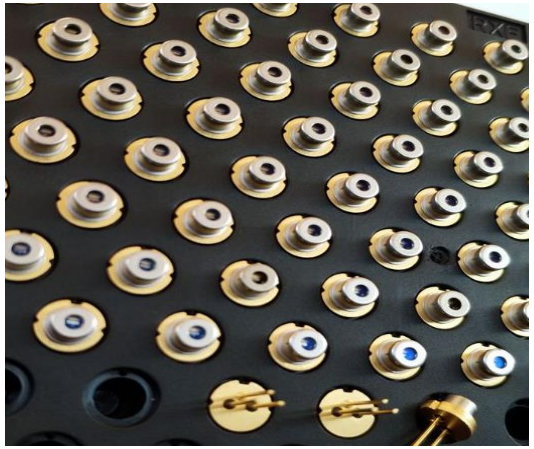
二极管：认识二极管



整流二极管



稳压二极管

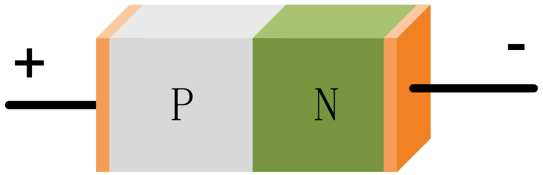


激光二极管

贴片二极管

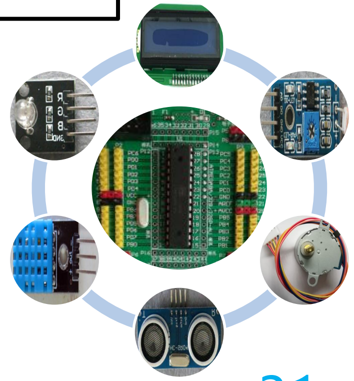
二极管的作用

半导体

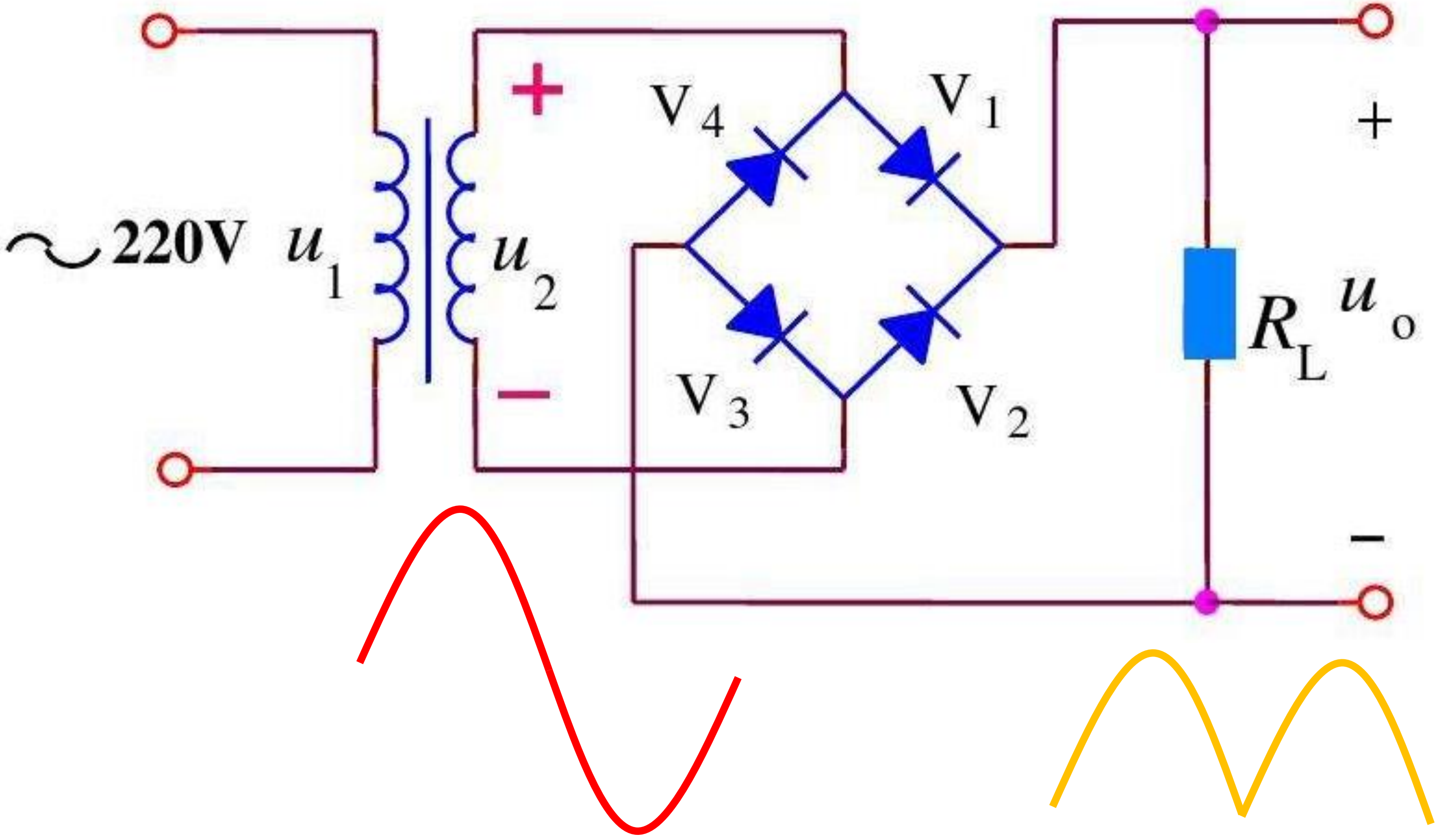


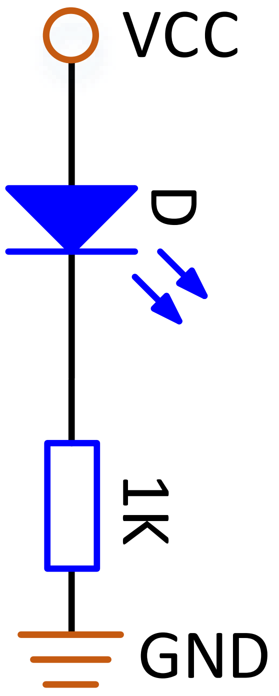
单向电流的传递

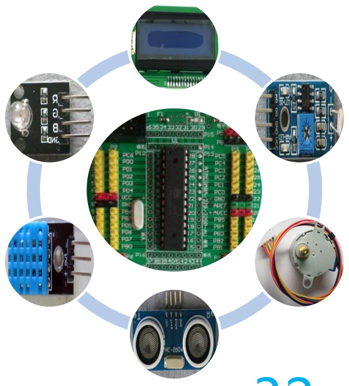
限幅、稳压、整流、调制、检波、光电转换…



二极管的作用







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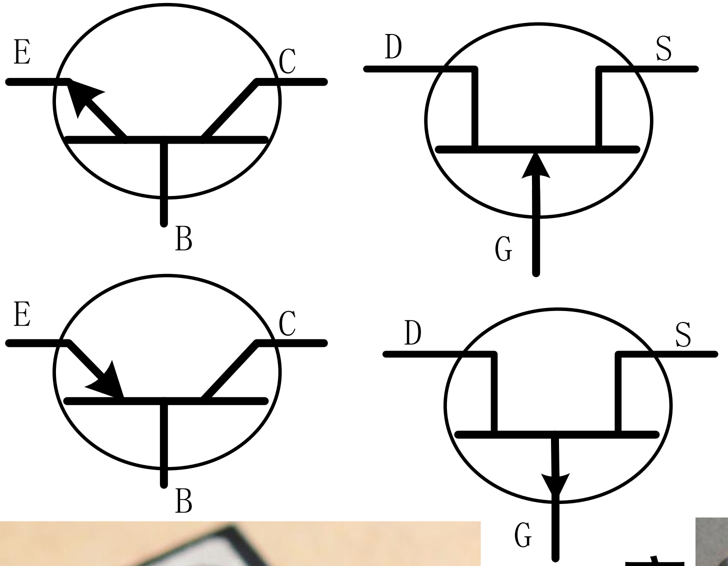
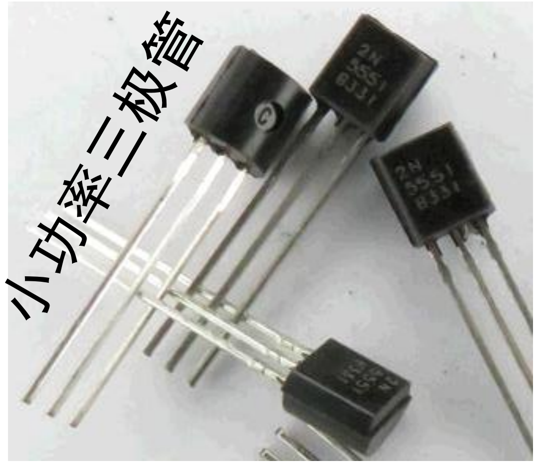
三极管：认识三极管

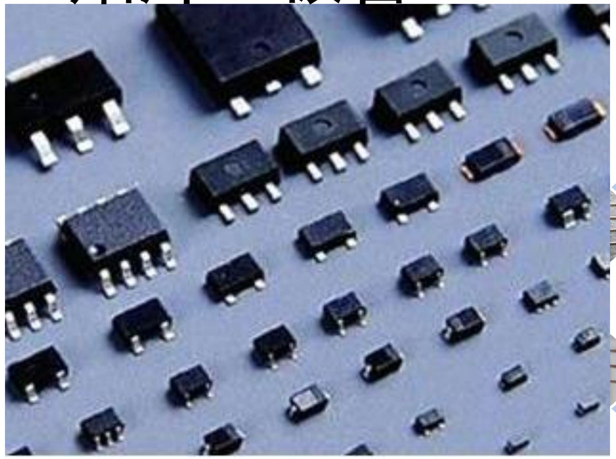


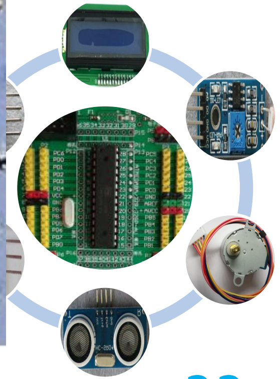


大功率三极管

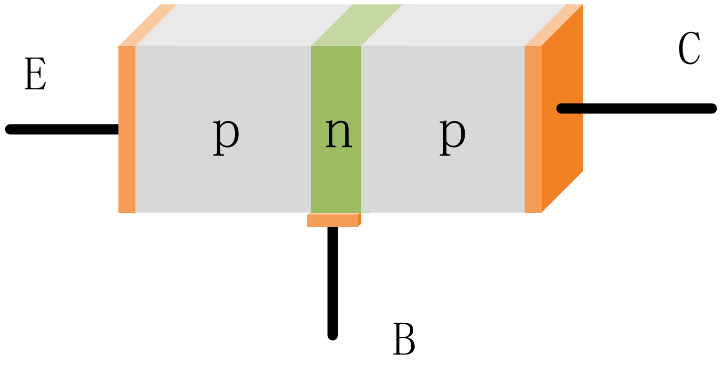
高频三极管

贴片三极管

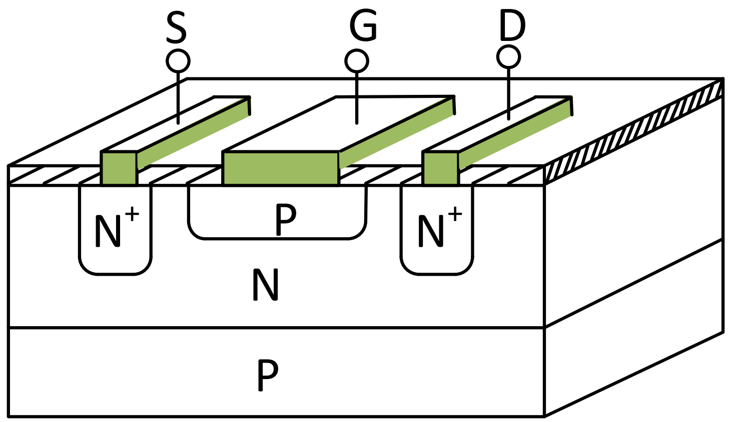




三极管的作用



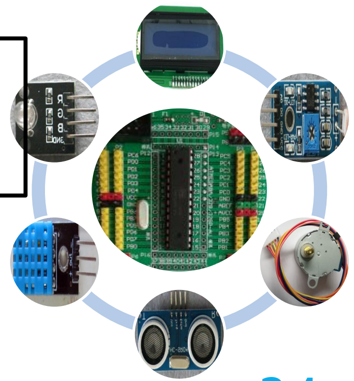
半导体



电流放大、电子开关

电子电路的核心

集成电路的基础



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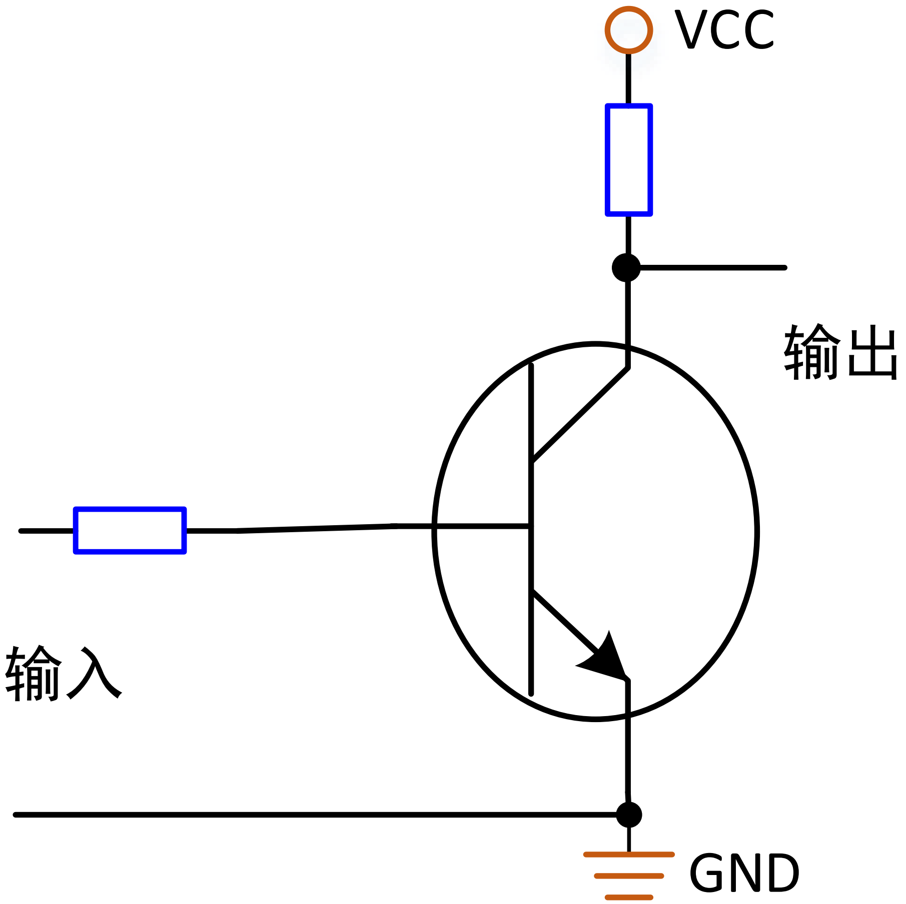
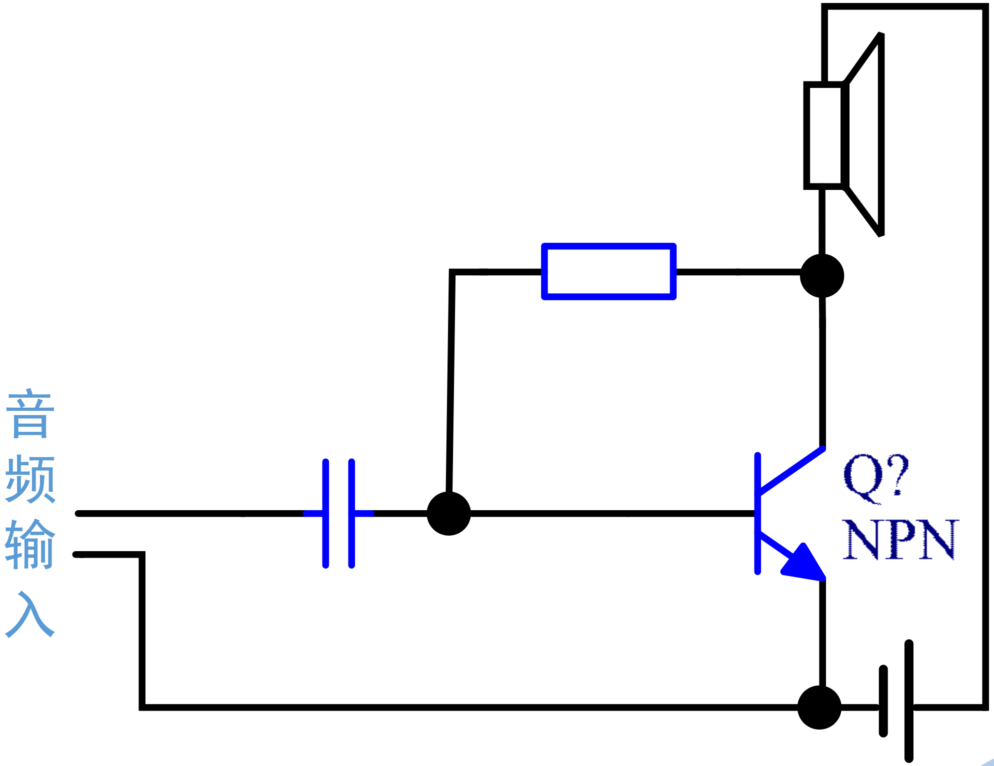
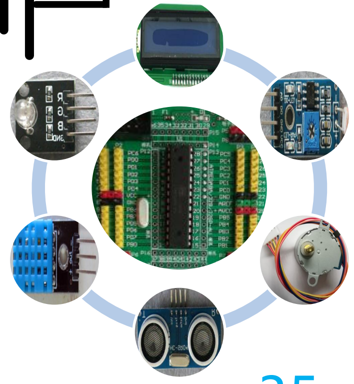
三极管的应用







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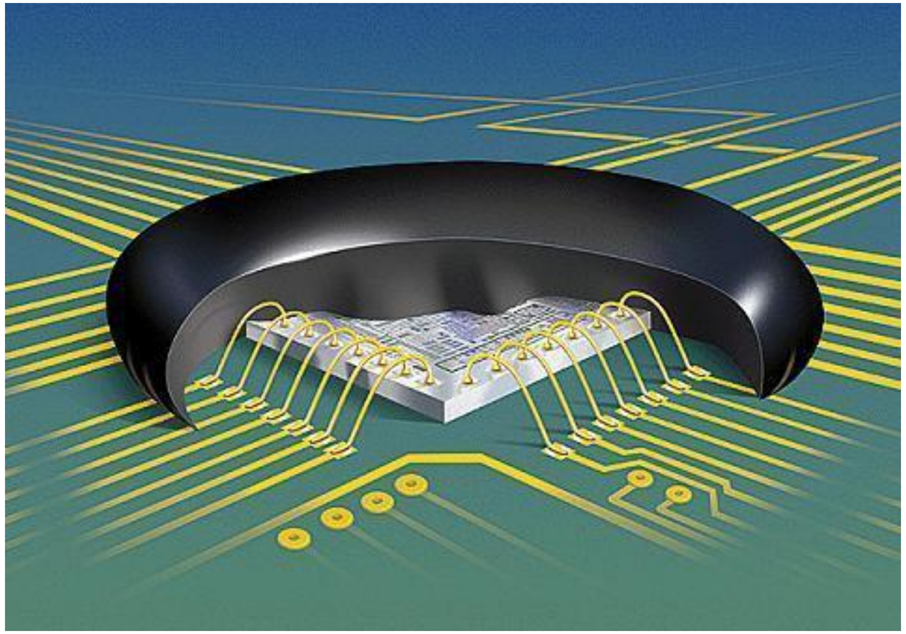
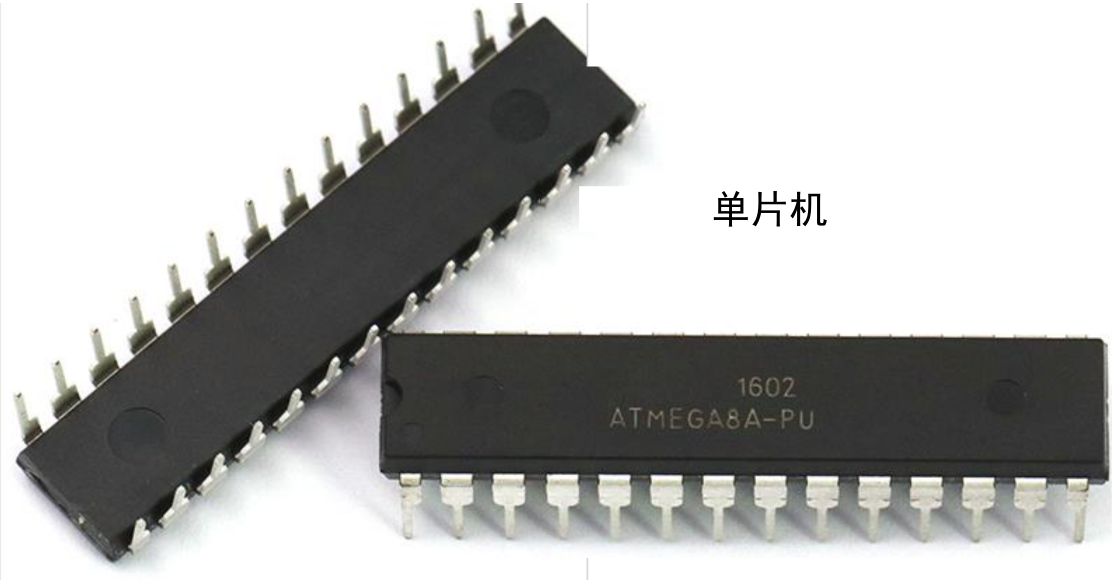
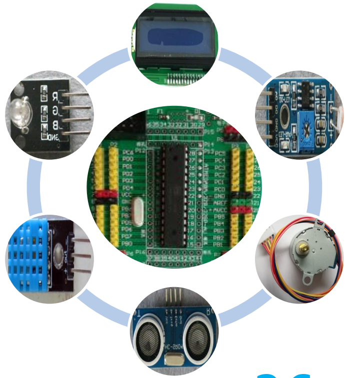
集成电路：认识1





单片机

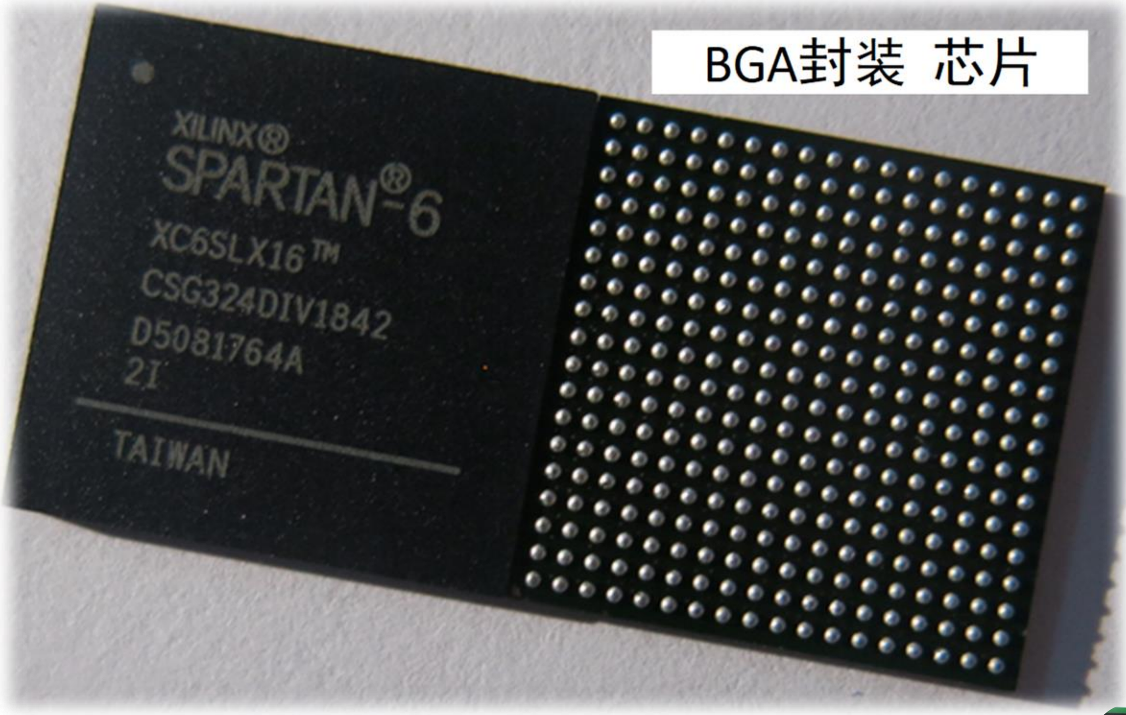
TQFP封装 芯片

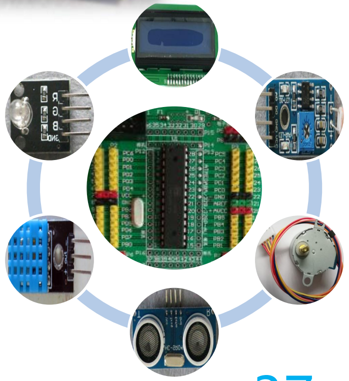


集成电路：认识2

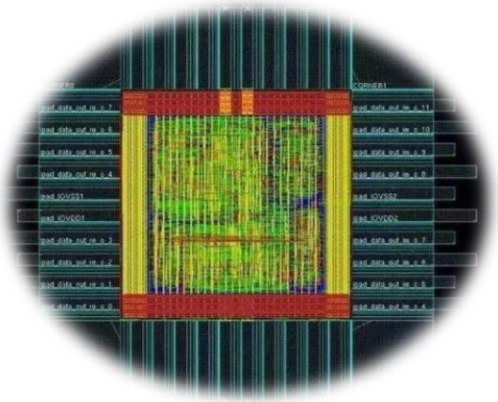
计算机CPU芯片

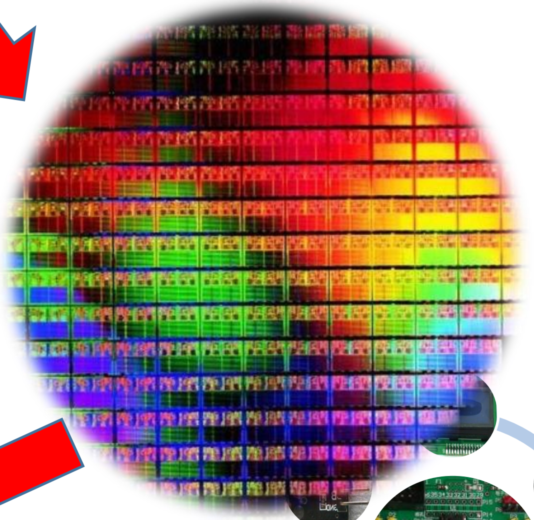
BGA封装 芯片

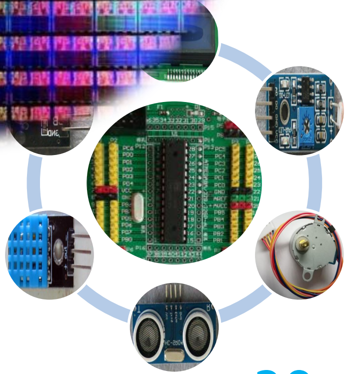


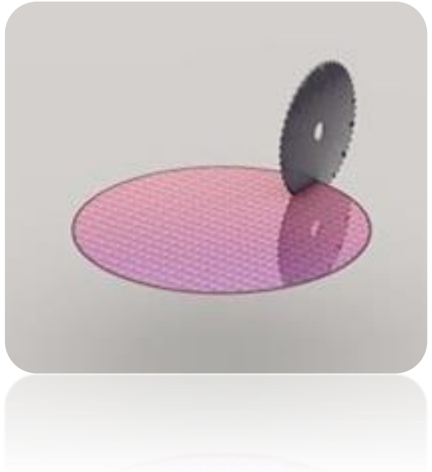


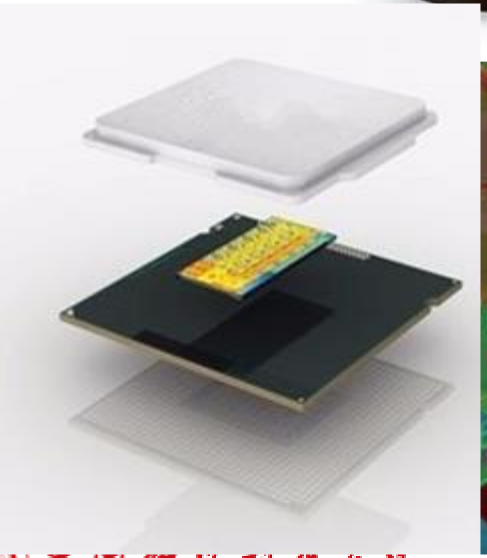
集成电路：制造

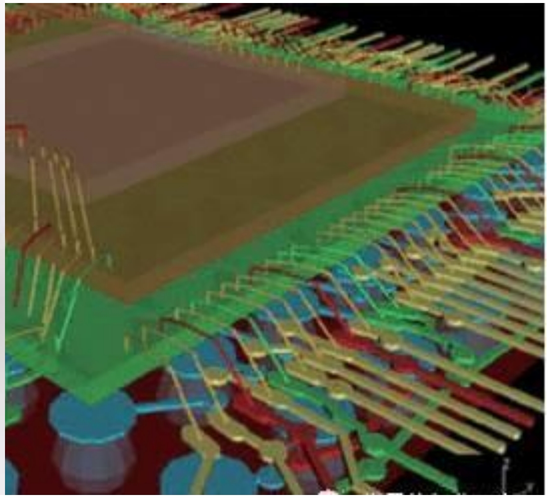








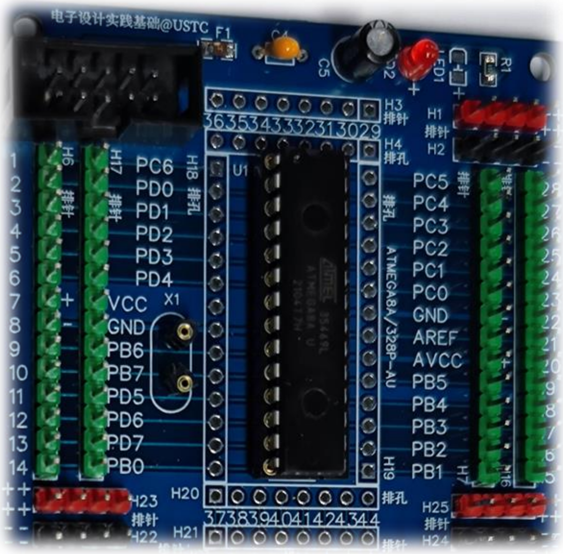




集成电路：使用

设计、
制造

?




```
int main(void)
{
    unsigned char sw_mode=0;
    unsigned int i,j,k;
    unsigned int de-
lay_num[10]={256,128,64,32,16,8,4,2,1,0};//
,60,40,30,20,15,10,9,8,7,6,5,3,2,1,1,1};
    _delay_ms(2000);//DHT11上电后须等待1秒
    //initialization
    DDRB = 0xc3; //11000011(数码管：PB7-4,PB
                    //模式选择(PB3,PB2)
    PORTB = 0x3c;//输入端口上拉

    DDRC = 0x27; //低3位设置为输
出:PC2(pin25),PC1(pin24),PC0(pin23)分别用于
G、B
```

集成电路：功能与应用

半导体芯片

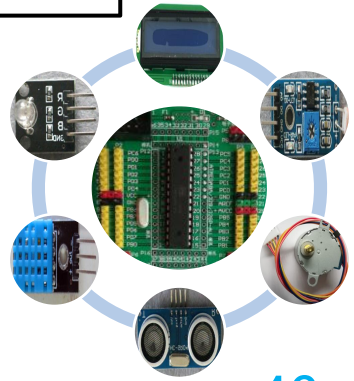
↓

功能、用途、规模

↙ ↘

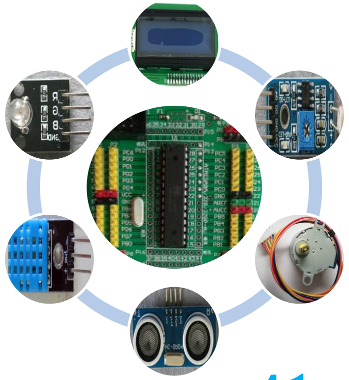
计算、存储、控制

通信、显示、安全



- 芯片在电子电路中的使用

几乎所有的电子设备或系统



谢 谢